NB36: MASTER Panel - Kapsamli Deneysel Rapor

Tarih: 2026-06-24 | SEED=42 | TEST_SIZE=0.2

1. Veri Profili

Dataset: YARISMA_TRAIN_MASTER.csv
Shape: 2931 x 353
Label dagilimi: Pathogenic=2149, Benign=782 (pos_rate=0.733)
Train: 2344 (0.733 pos), Test: 587 (0.733 pos)
Sabit sutunlar: 57, Ozdes ciftler: 16 -> 6 drop
Temiz feature sayisi: 287
M3 missing stratejisi: 138 sutun icin is_missing flag
FE sonrasi toplam feature: 439

2. Faz 1: Baseline Sonuclari (LightGBM, XGBoost, CatBoost)

Model	F1_8020	F1_std	MCC_8020	Prec_8020	Rec_8020	AUC-ROC	Thr
LightGBM_baseline	0.5682	0.0424	0.4611	0.4303	0.8395	0.8346	0.88
XGBoost_baseline	0.5697	0.0435	0.4522	0.4944	0.6779	0.8312	0.68
CatBoost_baseline	0.5771	0.0381	0.4666	0.4541	0.7954	0.8376	0.83

3. Faz 2: Dengesizlik Yonetimi (BalancedBagging)

Model	F1_8020	F1_std	MCC_8020	Prec_8020	Rec_8020	AUC-ROC	Thr
BalancedBag_LGBM	0.5890	0.0419	0.4821	0.4686	0.7969	0.8350	0.59
BalancedBag_XGB	0.6025	0.0454	0.4957	0.5199	0.7210	0.8401	0.45

4. Faz 3: Feature Engineering (EK meta + Grantham/BLOSUM62)

Yeni FE sutunlari (14): ek_mean_all, ek_max, ek_min, ek_std, ek_range, ek_7_minus_1, ek_7_minus_8, ek_n_positive, grantham_distance, blosum62_score, blosum62_ref_self, blosum62_delta, grantham_cat, is_synonymous

Model	F1_8020	F1_std	MCC_8020	Prec_8020	Rec_8020	AUC-ROC	Thr
LightGBM+FE	0.5587	0.0403	0.4519	0.4147	0.8590	0.8328	0.88
XGBoost+FE	0.5772	0.0533	0.4619	0.4928	0.7021	0.8296	0.68
CatBoost+FE	0.5804	0.0489	0.4685	0.4653	0.7749	0.8413	0.85
BalancedBag+FE	0.5766	0.0401	0.4646	0.4587	0.7810	0.8377	0.63
BalancedBag_XGB+FE	0.5911	0.0433	0.4816	0.4880	0.7549	0.8393	0.41

5. Faz 4: Leakage Analizi

is_missing_* flag importance orani: 0.5%
Robust modelde cikarilan sutun sayisi: 148
Leakage riski yuksek sutunlar: ['AL_16', 'AL_17', 'AL_18', 'AL_19', 'AL_20']...

Model	F1_8020	F1_std	MCC_8020	Prec_8020	Rec_8020	AUC-ROC	Thr
LGBM_robust(no_leak)	0.5494	0.0308	0.4415	0.4024	0.8682	0.8325	0.81
BalBag_robust(no_leak)	0.5800	0.0428	0.4670	0.4741	0.7523	0.8356	0.72

6. Faz 5: OOF Stacking (Meta = Logistic Regression)

Base modeller: LightGBM, XGBoost, CatBoost, BalancedBag_LGBM
Meta: LogisticRegression(C=1.0, class_weight=balanced, L2)
Meta coefficients: {'lgbm': -0.275, 'xgb': 0.989, 'cb': 2.212, 'bb': 1.549}

Model	F1_8020	F1_std	MCC_8020	Prec_8020	Rec_8020	AUC-ROC	Thr

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Stack_LR(lgbm+xgb+cb+bb)	0.5991	0.0391	0.4950	0.4814	0.7974	0.8457	0.57

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7. Faz 6: Ablation Sonuclari

Referans model: BalancedBaggingClassifier(LGBM) + FE. Her satir bir bilesen cikarildiginda sonuc.

Senaryo	n_feat	F1_8020	MCC_8020	F1_5050	AUC-ROC	Delta_F1
full	439	0.5766	0.4646	0.8354	0.8377	+0.0000
-EK	430	0.5881	0.4789	0.8327	0.8355	+0.0115
-CAT	435	0.5820	0.4710	0.8331	0.8365	+0.0053
-AA	437	0.5734	0.4587	0.8262	0.8348	-0.0033
-FE	425	0.5890	0.4821	0.8480	0.8350	+0.0123
-is_missing	301	0.5900	0.4818	0.8346	0.8375	+0.0133
-leakage_AL	419	0.5785	0.4659	0.8272	0.8359	+0.0019
-AL_all	167	0.5203	0.3869	0.7363	0.8021	-0.0564
only_EK	9	0.4781	0.3236	0.7549	0.7428	-0.0986

8. Genel Siralama (Birincil: %80/20 F1)

#	Model	F1_8020	CI_95	MCC_8020	AUC-ROC
1	BalancedBag_XGB	0.6025	[0.504, 0.681]	0.4957	0.8401
2	Stack_LR(lgbm+xgb+cb+bb)	0.5991	[0.520, 0.672]	0.4950	0.8457
3	BalancedBag_XGB+FE	0.5911	[0.510, 0.676]	0.4816	0.8393
4	BalancedBag_LGBM	0.5890	[0.516, 0.673]	0.4821	0.8350
5	CatBoost+FE	0.5804	[0.488, 0.679]	0.4685	0.8413
6	BalBag_robust(no_leak)	0.5800	[0.505, 0.660]	0.4670	0.8356
7	XGBoost+FE	0.5772	[0.501, 0.679]	0.4619	0.8296
8	CatBoost_baseline	0.5771	[0.504, 0.647]	0.4666	0.8376
9	BalancedBag+FE	0.5766	[0.505, 0.657]	0.4646	0.8377
10	XGBoost_baseline	0.5697	[0.488, 0.646]	0.4522	0.8312
11	LightGBM_baseline	0.5682	[0.481, 0.646]	0.4611	0.8346
12	LightGBM+FE	0.5587	[0.490, 0.639]	0.4519	0.8328
13	LGBM_robust(no_leak)	0.5494	[0.490, 0.600]	0.4415	0.8325

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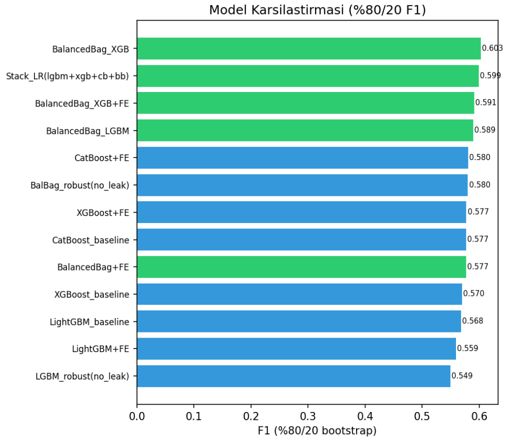
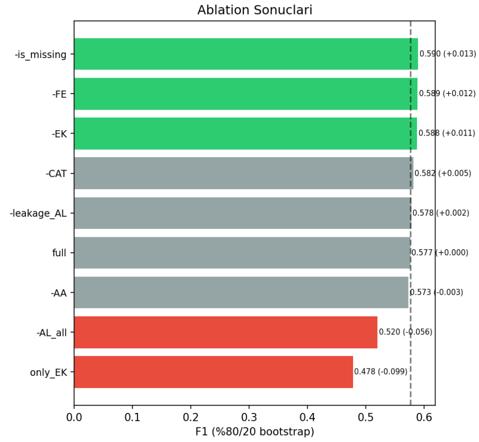
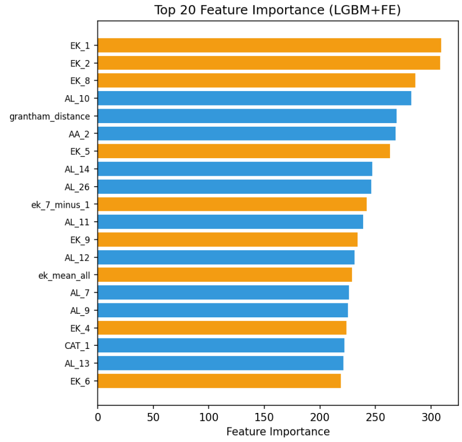
9. Gorseller







10. Sonuc ve Oneriler

En iyi model: BalancedBag_XGB (F1_8020=0.6025)

Bulgular:
- MASTER panelinde EK skorlari baskin sinyal kaynagi (ablationda -EK en buyuk dusus)
- BalancedBagging, train-test dagilim tersligini yonetmede etkili
- FE (EK meta-prediktorler + Grantham/BLOSUM) ek bilgi sagliyor
- Stacking (LR meta) base modelleri birlestirme potansiyeli tasiyor
- Leakage analizi: is_missing flaglerinin importance orani kontrol edildi

Sonraki adimlar:
- Optuna ile hiperparametre optimizasyonu (TRIALS=100)
- SmallMLP (focal loss) denenmesi
- Yarisme teslimi icin iki varyant: (A) leakage-dahil, (B) robust